VIT Bhopal University

B.Tech CSE (AI & ML)

Build Your Own Project (BYOP)

SymptoCare AI
AI-Powered Health Symptom Checker

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INTRODUCTION

SymptoCare AI is a Machine Learning-based health symptom checker developed to predict possible diseases using symptoms entered by users.

The project was created to explore the practical application of Machine Learning in healthcare-related problems. The system predicts diseases such as Flu, Dengue, COVID-19, and Typhoid using symptom-based analysis.

PROBLEM STATEMENT

Many people search random websites for health advice when experiencing symptoms, which can often lead to confusion or misinformation.

This project aims to create a simple AI-based system that predicts possible diseases from symptoms and provides basic precautionary suggestions.

OBJECTIVES

- To develop a disease prediction system using Machine Learning.
- To understand the implementation of classification algorithms.
- To create a simple and user-friendly healthcare application.
- To explore the use of AI in real-world problems.

TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- scikit-learn
- Streamlit
- Google Colab
- GitHub

MACHINE LEARNING MODEL

The project uses the Random Forest Classifier algorithm for disease prediction.

The model was trained using a symptom-based data set and predicts diseases based on user-selected symptoms.

WORKFLOW

User Symptoms
↓
Data Processing
↓
Machine Learning Model
↓
Disease Prediction
↓
Precaution Suggestion

IMPLEMENTATION

The dataset was created and processed using Pandas in Google Colab.

A Random Forest Classifier was trained using symptom data and saved using Joblib.

The trained model was integrated into a Streamlit-based interface where users can select symptoms and receive disease predictions instantly.

FEATURES

- Disease prediction system
- Symptom-based analysis
- Precaution recommendations
- Interactive user interface
- Fast prediction using Machine Learning

SCREENSHOTS

Dataset screenshot:-

...

1 to 10 of 10 entries Filter

index	fever	cough	headache	fatigue	vomiting	body_pain	disease
0	1	1	1	1	0	1	Flu
1	1	1	0	1	1	1	Dengue
2	0	1	1	0	0	0	Cold
3	1	0	1	1	1	1	Typhoid
4	1	1	1	1	1	1	COVID-19
5	0	0	1	1	0	1	Migraine
6	1	0	0	1	1	0	Malaria
7	0	1	0	0	0	0	Allergy
8	1	1	1	0	1	1	Pneumonia
9	0	0	1	1	0	0	Stress

Accuracy screenshot:-

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Features
X = df.drop("disease", axis=1)

# Target
y = df["disease"]

# Create model
model = RandomForestClassifier(n_estimators=100, random_state=42)

# TRAIN ON FULL DATASET
model.fit(X, y)

# Predict on same dataset
predictions = model.predict(X)

# Accuracy
accuracy = accuracy_score(y, predictions)

print("Accuracy:", accuracy * 100)
```

```
Accuracy: 100.0
```

Prediction output screenshot:-

```python
import pandas as pd

sample = pd.DataFrame([[
    1, 1, 1, 1, 0, 1
]], columns=[
    "fever",
    "cough",
    "headache",
    "fatigue",
    "vomiting",
    "body_pain"
])

prediction = model.predict(sample)

print("Predicted Disease:", prediction[0])
```

```
Predicted Disease: Flu
```

Live application screenshot:-

SymptoCare AI

AI-Powered Health Symptom Checker

Select the symptoms below to predict the possible disease using Machine Learning.

- [ ] Fever
- [ ] Cough
- [x] Headache
- [x] Fatigue
- [ ] Vomiting
- [ ] Body Pain

Predict Disease

Predicted Disease: Stress

Recommended Precaution: Practice relaxation and stress management.

⚠ This project is developed for educational purposes only and should not be considered professional medical advice.

CHALLENGES FACED

• Understanding Machine Learning workflow

• Managing dataset structure

• Integrating the ML model into the application

• Deploying the application online

FUTURE SCOPE

• Using larger healthcare datasets

• Adding chatbot support

• Voice-based symptom input

• Mobile application integration

CONCLUSION

SymptoCare AI demonstrates how Machine Learning can be used in healthcare-related applications for symptom-based disease prediction.

The project helped in understanding data processing, model training, deployment, and practical AI implementation.

REFERENCES

• Python Documentation

• scikit-learn Documentation

• Streamlit Documentation

• Pandas Documentation